MLflow — Experiment Tracking

Chapter 1 Test Questions | MLOps & Model Deployment

Level 1: Warmup Test • Level 2: Interviewer Test

🎯 LEVEL 1: WARMUP TEST

10 Questions | Build Your Foundational Understanding

SECTION A — Multiple Choice Questions (MCQ) [5 Questions]

Q1 [MCQ]

What is MLflow primarily described as in the context of machine learning workflows?

A. A deep learning framework for training neural networks

B. An open-source platform that acts as a 'lab notebook' for ML experiments, recording every run with its configuration and results

C. A cloud-based auto-scaling service for model inference

D. A data preprocessing pipeline tool built by Google

Q2 [MCQ]

Which MLflow component is responsible for storing structured data such as experiment names, run IDs, parameters, and metrics?

A. Artifact Store

B. MLflow UI

C. Backend Store (e.g., SQLite or PostgreSQL)

D. Tracking Client SDK

Q3 [MCQ]

Which MLflow function call enables automatic logging of parameters, metrics, and model artifacts without writing any manual log_*() calls?

A. mlflow.log_all()

B. mlflow.record_run()
C. mlflow.autolog()
D. mlflow.track_experiment()

Q4 [MCQ]
What is the key purpose of the `step` argument in mlflow.log_metric('val_loss', 0.45, step=epoch)?

A. It sets the delay (in seconds) before the metric is saved to the server
B. It specifies which database table to write the metric value into
C. It records the metric at each epoch so training curves (time-series charts) are visible in the MLflow UI
D. It increases the logging precision to avoid floating-point rounding

Q5 [MCQ]
What is the correct command to start an MLflow tracking server using a local SQLite database as the backend store?

A. mlflow run --db sqlite:///mlflow.db
B. mlflow server --backend-store-uri sqlite:///mlflow.db --default-artifact-root ./mlartifacts --host 0.0.0.0 --port 5000
C. mlflow start --backend sqlite --port 5000
D. python -m mlflow.server --uri sqlite --port 5000

🕮 SECTION B — Descriptive Questions [3 Questions]

Q6 [DESCRIPTIVE]
Describe the four core components of MLflow's architecture and explain the role each plays in the experiment tracking lifecycle.

ANS

Q7 [DESCRIPTIVE]

Why does experiment tracking matter in MLOps? Explain at least four concrete consequences of not having it in a production ML team.

ANS

Q8 [DESCRIPTIVE]

What is the difference between 'parameters' and 'metrics' in MLflow? Why does MLflow treat them differently, and what API calls correspond to each?

ANS

SECTION C — Coding Questions [2 Questions]

Q9 [CODING]

Write a complete MLflow experiment tracking script that: (1) sets the tracking URI and experiment name, (2) logs at least 4 hyperparameters, (3) trains a RandomForestClassifier on the breast cancer dataset, (4) logs 3 evaluation metrics, and (5) logs the trained model as an artifact.

ANS



ANS

Q10 [CODING]

Enable MLflow auto-logging for scikit-learn, train a LogisticRegression model on the Iris dataset, and add two custom tags and one custom business metric on top of the auto-logged data.

ANS **Auto-logging is powerful — here's how to use it while still adding your own custom context:**

ANS

🚀 LEVEL 2: INTERVIEWER TEST

10 Intermediate–Semi-Advanced Questions | Think Like a Senior Practitioner

🎭 SECTION A — Scenario-Based MCQ [3 Questions]

Q1 [SCENARIO MCQ]

SCENARIO: A fintech startup has 2 data scientists who started with local MLflow (./mlruns folder). The team has grown to 8 scientists, and now no one can see each other's experiments — everyone has a separate local database. What is the CORRECT next step to fix this?

A. Ask each data scientist to manually copy their mlruns folder to a shared network drive every evening

B. Deploy a shared MLflow tracking server on a central machine (or EC2), configure a PostgreSQL backend store, set up an S3 artifact store, and have all team members update their mlflow.set_tracking_uri() to point to the shared server URL

C. Switch from MLflow to Weights & Biases because MLflow does not support multi-user setups

D. Create a separate Experiment for each data scientist so their runs don't interfere

Q2 [SCENARIO MCQ]

SCENARIO: After 60 training runs over 3 weeks, a data scientist tells their manager 'Run 47 was the best — 93% AUC-ROC.' The manager asks them to redeploy that model. The scientist can't — they have no record of Run 47's hyperparameters, and the model file is gone. Which MLflow features, if used from the start, would have prevented this?

A. mlflow.set_experiment() and mlflow.set_tag() only

B. mlflow.log_params() to record hyperparameters AND mlflow.sklearn.log_model() (or log_artifact) to permanently link the model file to the run — both retrievable via the run's unique run_id at any time

C. mlflow.search_runs() with a filter_string — it can reconstruct lost runs from the database

D. mlflow.autolog(disable=True) to prevent accidental overwriting of run data

Q3 [SCENARIO MCQ]

SCENARIO: A healthcare AI team is building a diagnostic model. Regulators require that for any model decision, the team must produce: (a) exact model artifact, (b)

preprocessing pipeline, (c) training data statistics, and (d) validation plots — all from the same training run. Which MLflow approach BEST satisfies this compliance requirement?

A. Log only the final model using mlflow.sklearn.log_model() and keep other files in a separate S3 bucket

B. Use mlflow.log_artifact() individually for the model file only — other files can be stored in git

C. Log all four items as artifacts in the same MLflow run using mlflow.sklearn.log_model() for the full pipeline, mlflow.log_artifact() for the data statistics CSV and validation plots — all linked to one run_id retrievable at any future date

D. Store everything in a relational database and only use MLflow for metric tracking

SECTION B — Scenario-Based Descriptive [2 Questions]

Q4 [SCENARIO DESCRIPTIVE]

SCENARIO: A 3-person ML startup is expanding to 8 data scientists. They currently run mlflow locally (SQLite backend, local artifacts). Describe step-by-step how you would migrate this setup to support the full team, and explain why each step is necessary.

ANS

Q5 [SCENARIO DESCRIPTIVE]

SCENARIO: A hospital AI team needs their experiments to reproduce identically across (1) data scientists' laptops, (2) a GPU training cluster, and (3) a CI/CD pipeline — while remaining HIPAA-compliant. Design the MLflow setup to satisfy these requirements.

ANS

SECTION C — Coding Questions [3 Questions]

Q6 [CODING]

Write a training loop that logs train_loss, val_loss, train_acc, and val_acc at EVERY epoch using the step parameter, so all four metrics appear as time-series charts in the MLflow UI. Use 15 simulated epochs.

ANS

The step parameter is what transforms a flat metric into a living training curve. Here's the full implementation:

ANS

Q7 [CODING]

Write a script that logs a FULL sklearn pipeline (preprocessor + GradientBoostingClassifier) as a single MLflow artifact, then shows how to retrieve and use that pipeline in a separate prediction script.

ANS

Logging only the model without the preprocessing pipeline is one of the most common MLOps mistakes — it breaks deployment. Here's the correct approach:

ANS

Q8 [CODING]

Write a Python script that uses mlflow.search_runs() to find the top 5 runs from a specific experiment filtered by AUC-ROC > 0.88, sorted by AUC-ROC descending, and then programmatically loads and uses the best model for predictions.

ANS

This is the programmatic version of what you'd normally do in the UI — essential for CI/CD pipelines that need to auto-select the best model:

ANS

SECTION D — Database Scenario-Based Coding [2 Questions]

Q9 [DB CODING]

SCENARIO: Your team is moving MLflow to production. Write the complete setup — including the Docker Compose configuration for a PostgreSQL backend store and the mlflow server startup command — so the entire team can share a single, persistent tracking server.

ANS **This is the production setup pattern from section 1.3 expanded with explanation:**

ANS

Q10 [DB CODING]

SCENARIO: Your MLflow PostgreSQL backend has 300+ runs across 5 experiments. Write Python code that: (1) queries all runs where f1_score > 0.85 AND n_estimators > 100, (2) groups results by experiment name and shows count + avg AUC per group, and (3) exports the filtered results to a CSV report.

ANS **This is a real-world data analysis task you'd do before a model review meeting. Here's the complete solution:**

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